

Conditional Statements in Python

* Conditional Statements helps a program make decisions

* Sometimes we want code to run only when a condition is true

eg:

⇒ checking if a student passed

⇒ checking if person can vote

i. The if Statement:

* The if statement runs code only when the condition is True

Syntax:

if condition:

statement

Note:

* : ⇒ (colon) is required

* code inside if must be indented

ii. The else Statement:

* Sometimes we want the program to run another block of code when the condition is False

Syntax:

if condition:

statement

else:

statement

iii. The elif Statement:

* Sometimes we need to check multiple conditions

* Instead of writing many if statements, Python use ~~else~~ elif.

Syntax:

if condition1:

statement

elif condition2:

statement

else:

statement

Note:

Python stops checking conditions after the first True condition.

* Even though the second condition is true, Python does not check it

Nested-if statement

- * A nested if statement meaning placing an if statement inside another if statement
- * It is used when one condition depends on another condition

Syntax:

if condition 1:

if condition 2:

statement

Note:

Python uses indentation to understand nested blocks

Quiz:

1. Write a program that checks

* if marks $\geq 50 \rightarrow$ student passed

* if marks $\geq 80 \rightarrow$ student gets distinction

2. When the inner if statement executes

a. Always

b. When the first condition is True

c. When the first condition is False

d. Never